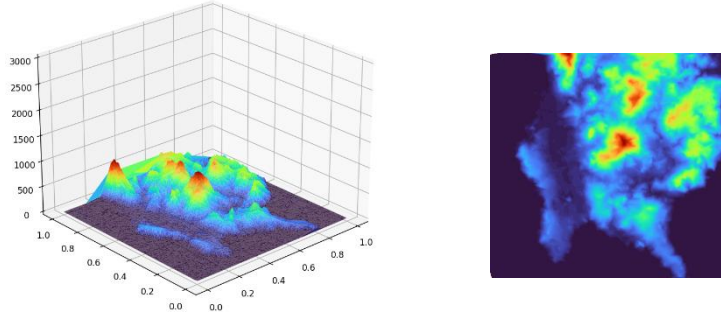


Machine Learning (2025)

Homework 1: Linear Regression

Deadline: 2025/3/26

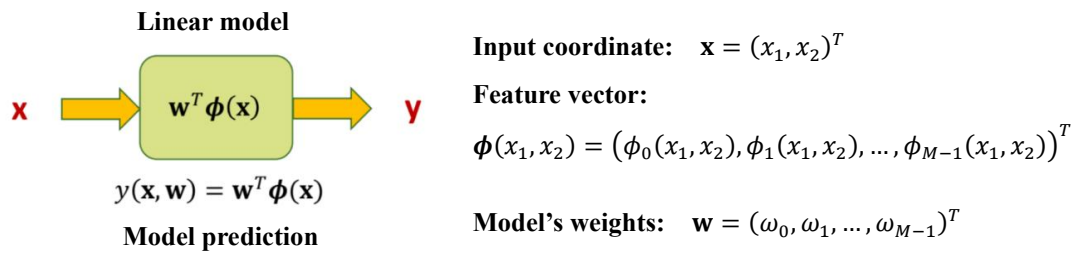
In this homework, you are required to reconstruct the altitude data of southern Taiwan using three regression approaches based on the given training data.



The coordinate values in this area have been normalized to the range $[0.0, 1.0] \times [0.0, 1.0]$, and the altitude values are **nonnegative**. You will receive a training dataset containing **6,000** pairs of coordinates and their corresponding altitude values.

Then, create feature vectors based on the input coordinates as the model's inputs and train your regression models to fit the training data.

Finally, predict the altitude values for the **2,000** coordinates in the testing dataset and **save your predictions along with the model's weights** as a .csv file. (The TA will run your programs, so just ensure that the output format will be correct.)



For each approach, the mean squared error (MSE) of your predictions on the testing dataset must be **less than 900**; otherwise, you will fail the correctness check.

$$\text{MSE} = \frac{1}{N} \sum_i (t_i - y(\mathbf{x}_i, \mathbf{w}))^2$$

where $y(\mathbf{x}_i, \mathbf{w})$ is your predicted value, and t_i is the ground truth.

$$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w})}{p(\mathcal{D})} \propto p(\mathcal{D}|\mathbf{w})p(\mathbf{w})$$

$p(\mathbf{w})$: Prior probability

$p(\mathcal{D}|\mathbf{w})$: Likelihood function of $\mathbf{w}$

$p(\mathbf{w}|\mathcal{D})$: Posterior probability

Likelihood	Prior	Name
Gaussian	Uniform	Least Squares
Laplace	Uniform	Robust Regression
Student	Uniform	Robust Regression
Gaussian	Gaussian	Ridge Regression
Gaussian	Laplace	Lasso Regression

1. Maximum Likelihood (ML) Approach (20%)

Please select $p(\mathcal{D} | \mathbf{w})$ and $\phi_j(x_1, x_2)$ to implement your ML regression approach.

[Input] training_dataset.csv, testing_dataset.csv

[Output] result_1.csv

For example, if you choose $p(\mathcal{D} | \mathbf{w})$ to be Gaussian distribution, then the objective function is

$$E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{t_n - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_n)\}^2$$

Next, if you uniformly place N Gaussian basis functions, they are defined as

$$\phi_j(x_1, x_2) = \exp\left(-\frac{(x_1 - \mu_{j_1})^2}{2\sigma_{j_1}} - \frac{(x_2 - \mu_{j_2})^2}{2\sigma_{j_2}}\right), \quad \text{for } 1 \leq j_1 \leq O_1, 1 \leq j_2 \leq O_2$$

The index j is given by

$$1 \leq j = j_1 + O_2(j_2 - 1) \leq N, \quad N = O_1 O_2$$

2. Maximum A Posterior (MAP) Approach (20%)

Similar to the ML approach, but with an additional prior term (i.e., adding a regularization term).

[Input] training_dataset.csv, testing_dataset.csv

[Output] result_2.csv

3. Bayesian Linear Regression (20%)

Please use the formula on p.51 of the Chapter 3 (Regression) handout to implement Bayesian linear regression.

$$p(t|\mathbf{x}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(t | \mathbf{m}_N^T \boldsymbol{\phi}(\mathbf{x}), \sigma_N^2(\mathbf{x}))$$

$$\mathbf{m}_N = \mathbf{S}_N(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}^T\mathbf{t}), \quad \mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^T\boldsymbol{\Phi}, \quad p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0)$$

Since the prediction follows a distribution, you can output the **mean value** as the final result. In this part, save $\mathbf{m}_N$ as the model's weights in the output file.

[Input] training_dataset.csv, testing_dataset.csv

[Output] result_3.csv

Hints:

- You may assume that the prior covariance matrix $\mathbf{S}_0 = \alpha^{-1}\mathbf{I}$, where α is a hyperparameter.

4. Performance Ranking (10%)

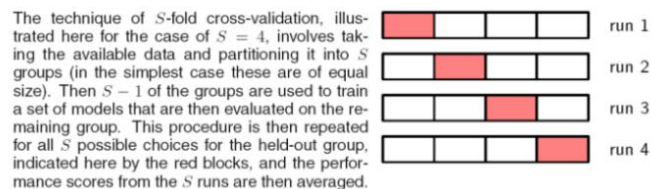
After passing the correctness check, the TA will evaluate your performance based on the results of your three approaches and select the best one for ranking. Your performance score will be calculated using the following formula:

$$\text{Performance Score} = \text{Mean Squared Error} \times \text{Number of Weights}$$

Your final score in this part will be determined based on the ranking of your performance score. (The lower performance score, the higher final score.)

S-Fold Cross-Validation:

Since you will only receive the training dataset, you can use this cross-validation technique to evaluate your model's accuracy.



This method provides a robust estimate of the performance of the three regression models and helps you determine the best hyperparameters.

Provided / Output .csv Files:

Below is the information about the provided files and expected output files.

training_dataset.csv

	A	B	C
1	0.092131	0.937033	0
2	0.125278	0.488667	0
3	0.652849	0.907492	318
4	0.812601	0.433788	158
5	0.850322	0.715338	336
6	0.460064	0.834544	62
7	0.704118	0.322039	129
8	0.283906	0.895476	52
9	0.126323	0.604167	9
10	0.279143	0.987275	320
11	0.842878	0.245481	0
12	0.358752	0.72852	177
13	0.39845	0.039886	0
14	0.165689	0.961357	0
15	0.370403	0.149431	0
16	0.217106	0.636854	31
17	0.060973	0.130356	0
18	0.988887	0.598953	0
19	0.317704	0.836527	233
20	0.323919	0.928939	363
21	0.588042	0.734457	364
22	0.024924	0.923444	0
23	0.279791	0.036062	0

training_dataset

coordinates

altitudes

testing_dataset.csv

	A	B	C
1	0.584704	0.5533	
2	0.435882	0.370251	
3	0.698013	0.090182	
4	0.626066	0.140503	
5	0.631488	0.381348	
6	0.537908	0.068881	
7	0.522531	0.284556	
8	0.969811	0.343638	
9	0.737205	0.756141	
10	0.550423	0.056492	
11	0.373115	0.940169	
12	0.65001	0.769146	
13	0.120348	0.119319	
14	0.200229	0.324696	
15	0.195694	0.308918	
16	0.432626	0.847454	
17	0.490827	0.032677	
18	0.432096	0.261818	
19	0.544423	0.768849	
20	0.250177	0.328927	
21	0.34757	0.180109	
22	0.82891	0.387402	
23	0.533625	0.003294	

testing_dataset

coordinates

result_1/2/3.csv

	A	B	C
1	251.5839	-5.63777	
2	193.8572	17.88713	
3	36.1378	16448.44	
4	27.18023	-165.133	
5	76.77199	119.7207	
6	0.353599	66.47781	
7	116.6401	2288.4	
8	0	70.37046	
9	234.8341	69.36864	
10	0.701127	-39.6392	
11	119.1363	-1439.08	
12	210.1447	17.45374	
13	1.101557	31.01605	
14	0	143.5233	
15	0	62.55628	
16	60.15256	-322.232	
17	0	988.9483	
18	57.63961	-33.9595	
19	224.1996	657.2062	
20	57.8264	6446.175	
21	14.72703	-38.472	
22	10.14655	-21.1096	
23	0	-123.186	

result_1

altitudes

weights

Please refer to the following code to save your results:

```
import numpy as np

SAVE_PATH = "outputs/result_1.csv"

def save_result(preds: np.ndarray, weights: np.ndarray):
    """Save prediction and weights to a CSV file
    - `preds`: predicted values with shape (n_samples,)
    - `weights`: model weights with shape (n_basis,)
    """
    max_length = max(len(preds), len(weights))

    result = np.full((max_length, 2), "", dtype=object)
    result[:len(preds), 0] = preds.astype(str)
    result[:len(weights), 1] = weights.astype(str)

    np.savetxt(SAVE_PATH, result, delimiter=",", fmt="%s")
```

Please refer to the following code to read data from “training_dataset.csv” and “testing_dataset.csv”:

```
import numpy as np
import pandas as pd

# Files path setting
TRAIN_PATH = "inputs/training_dataset.csv"
TEST_PATH = "inputs/testing_dataset.csv"

# Load data
train_set = pd.read_csv(TRAIN_PATH, header=None).values.astype(np.float32)
test_set = pd.read_csv(TEST_PATH, header=None).values.astype(np.float32)
```

Homework Rules and Grading Policy

Homework will be graded by:

1. Program Correctness (60%)

For each approach, the mean squared error (MSE) of your predictions on the testing dataset must be less than 900. (TA will provide an additional small testing dataset to help you verify that your output format is correct.)

2. Performance (10%)

3. Report (30%)

- Maximum Likelihood Approach
- Maximum A Posterior Approach (e.g., likelihood, basis functions, prior)
- Bayesian Linear Regression
- Additional Details (e.g., preprocessing, comparison)

Remind:

- **Program:** Please use **Python** and only the **NumPy** package for regression models. Since altitude values are always nonnegative, you may set negative predicted values to 0 to improve your performance score.
- **Report:** Use font size **12**, with a maximum of **7** pages. Explaining the key aspects of your regression approaches and visualizing your experimental results will earn a higher score on the report.
- **Performance:** Failure to pass all correctness check will result in a score of zero.
- **Deadline:** Late submissions within **3** days receive 70% of the score. Submissions beyond **3** days are NOT accepted.
- **Test Failure:** If you do not pass the correctness check, half of the score will be deducted, but there is a chance for remediation.
- **Plagiarism:** If two codes show high similarity, both will receive zero marks after confirmation.

Upload:

[Web] E3

[File Name] hw1_{student id}.zip

(e.g., hw1_123456789.zip)

```
hw1_{student id}/
├── inputs/
│   ├── training_dataset.csv
│   └── (testing_dataset.csv)
├── outputs/
│   ├── (result_1.csv)
│   ├── (result_2.csv)
│   └── (result_3.csv)
├── report.pdf
├── problem_1.py
├── problem_2.py
├── problem_3.py
└── README.txt
```